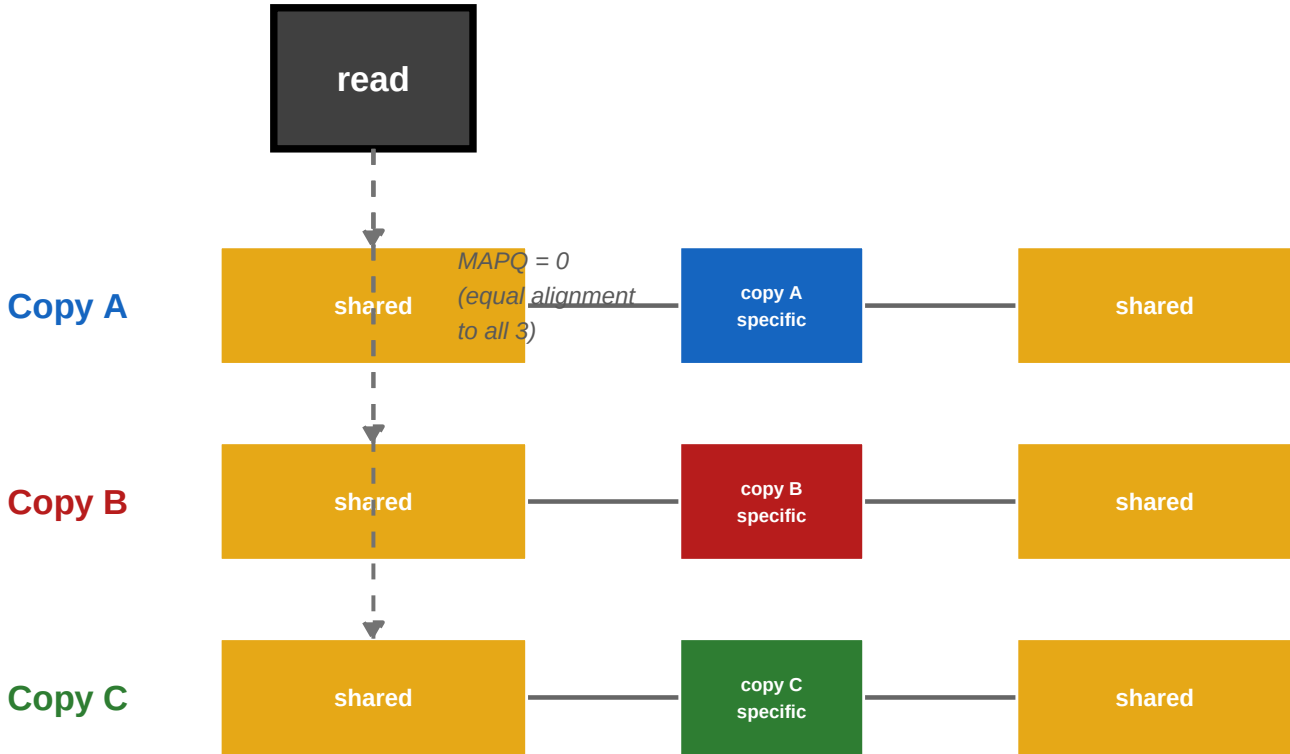


Per-copy profile HMMs resolve multimapping reads without diagnostic SNPs

Each copy's HMM scores the full read alignment. Small per-position differences accumulate across ~100 bp to distinguish copies.

A Multi-mapping read: no diagnostic position, equal MAPQ to all 3 loci



B Per-copy profile HMM scores

$DlogL = \log P(\text{read} | \text{HMM}_i) - \log P(\text{read} | \text{best HMM})$
No SNP required – cumulative signal over full read

